1 Groups

We start by recalling the definition of a group and reviewing some examples that you saw last year.

Definition 1.1 (Group).
A group is a pair $(G, .)$ where G is a non-empty set and . is a composition law

$$(.) : G \times G \to G$$

$$(a, b) \mapsto a.b$$

which satisfies the following conditions:

1. $(.)$ is associative: $(a.b).c = a.(b.c)$.
2. $(.)$ admits a neutral element e_G (simply denoted e if there is no risk of confusion) satisfying:

$$a.e_G = e_G.a = a, \ \forall a \in G.$$

3. existence of an inverse: $\forall a \in G, \exists a^{-1} \in G$:

$$a.a^{-1} = a^{-1}.a = e_G,$$

a^{-1} is called the inverse of a.

If the composition law $(.)$ further satisfies $a.b = b.a$ for all $a, b \in G$, we say G is *abelian* or commutative. We say G is *finite* if it has finitely many elements. The number of elements of G is denoted $|G|$ and called the *cardinality* or *order*[1] of G. If G is not finite we say it is *infinite*.

Example 1.1.

1. $(\mathbb{Z}, +)$ the set of integers endowed with the addition law is an abelian, infinite, group. The neutral element is 0 and the inverse of an integer $n \in \mathbb{Z}$ is $-n$.
2. Given $n \geq 1$, recall the integers modulo n:

$$\mathbf{Z}_n = \{[0]_n, [1]_n, \cdots, [n-1]_n\}$$

where $[k]_n$ is the class of the integer k modulo n, $[k]_n = [l]_n$ if and only if $k - l$ is divisible by n.
The addition of integers modulo n is defined by $[i]_n + [j]_n = [i+j]_n$. The set of integers modulo n endowed with the addition law $(\mathbf{Z}_n, +)$ is a finite abelian group of cardinality $|\mathbf{Z}_n| = n$. The neutral element is $[0]_n$, and the inverse of $[k]_n \in \mathbf{Z}_n$ is $[-k]_n$.
3. Given $n \geq 1$, $(S_n, \circ)$ is the group of permutations of the set $A = \{1, \ldots, n\}$: a permutation of A is a bijection $A \to A$. An element $\sigma \in S_n$ is a bijective map $\sigma : A \to A$.
The composition law $\circ$ is the composition of maps: if $\sigma, \tau \in S_n$ then

$$\sigma \circ \tau : A \to A$$

is defined by

$$(\sigma \circ \tau)(a) = \sigma(\tau(a)) \qquad \forall a \in A.$$

[1] Do not confuse this with the order of an element of a group

The neutral element $e = e_{S_n}$ is the identity map $\mathrm{id}_A : A \to A$ (it is a bijection).
S_n is a finite group with $|S_n| = n!$.
Assume $n \geq 2$, and $2 \leq k \leq n$ is an integer. Recall $\sigma = (a_1 a_2 \dots a_k)$ is a *cycle* if $\{a_1, a_2, \dots, a_k\} \subset A$ are k distinct elements of A, $\sigma(a_i) = a_{i+1}$ if $1 \leq i \leq k-1$, $\sigma(a_k) = a_1$, and $\sigma(j) = j$ if $j \notin \{a_1, a_2, \dots, a_k\}$. The integer k is called the *length* of σ.
The number of cycles of length $2 \leq k \leq n$ in S_n equals $\frac{n(n-1)\cdots(n-k+1)}{k}$ (prove this fact).
S_n is not abelian if $n \geq 3$: $(123)(13) \neq (13)(123)$.

4. For $n \geq 3$, $(D_{2n}, .)$ is the group of rotational and reflection symmetries of a regular n-gon Γ_n whose vertices we label by $1, 2, \dots, n$. Let r be the rotation clockwise in the plane of Γ_n with angle $\frac{2\pi}{n}$, and s the rotation through the symmetry axis of Γ_n passing through the vertex 1. Then
$$D_{2n} = \{1, r, \dots, r^{n-1}, s, sr, \cdots, sr^{n-1}\}$$
has $2n$ elements, 1 is the identity rotation, $r^n = s^2 = 1$, $(r^i)^{-1} = r^{-i} = r^{n-i}$, and $sr^i = r^{-i}s = r^{n-i}s$. The group D_{2n} is not abelian if $n \geq 3$: $sr \neq rs = sr^{n-1}$ in this case since $r \neq r^{n-1}$.
Here $r^k = r \circ \dots \circ r$ means r composed with itself k times for $k \geq 1$, it is the rotation clockwise in the plane of Γ_n with angle $\frac{2\pi k}{n}$.
For $n = 2$ we define D_4 as follows
$$D_4 = \{1, s, r, sr\}$$
with 1 the neutral element, $s^2 = r^2 = 1$, and $sr = rs$. So D_4 is abelian. Note that $(sr)^2 = srsr = srrs = sr^2s = ss = s^2 = 1$ in this case.

Remark (Basic Properties).
Let G be a group with neutral element e, and $a, b \in G$.

1. The neutral element e is unique.
2. The inverse a^{-1} of an element a is unique.
3. $(a^{-1})^{-1} = a$.
4. $(ab)^{-1} = b^{-1}a^{-1}$.
5. Cancellation laws: $\{au = av\} \Longrightarrow \{u = v\}$ and $\{ub = vb\} \Longrightarrow \{u = v\}$.
6. For $n \geq 1$, $a^n = a.a \dots a$ is the composite of a with itself n times.
If $n < 0$ we define $a^n = (a^{-n})^{-1}$.
Exponentiation rules: we have $a^n.a^m = a^{n+m}$, and $(a^n)^m = a^{nm}$, for $n, m \in \mathbb{Z}$, and $a \in A$.
7. The associativity of the group law (.) implies that for $a_1, \dots, a_k \in G$ the product $a_1.a_2.\dots.a_k$ is well defined and is independent of the way the expression is bracketed, however the order of the a_i's in the composition process is important unless G is abelian.

Definition 1.2 (Subgroup).
Let $(G, .)$ be a group. A *subgroup* of G is a non-empty subset $H \subset G$ which is closed under the composition law (.) and taking inverses:

- $x, y \in H \quad \Longrightarrow \quad x.y \in H$,
- $x \in H \quad \Longrightarrow \quad x^{-1} \in H$.
In particular if $x \in H$ then $x.x^{-1} = e_G \in H$.
In other words, H is a subgroup of G if H is closed under the composition law (.), and endowed with the latter $(H, .)$ is a group on its own.

Example 1.2.
Check the following are subgroups.

1. Given a group G the subset $\{e_G\}$ is a subgroup called the trivial subgroup.
2. Given $m \in \mathbb{Z}$ the subset $m\mathbb{Z} = \{mk \ : \ k \in \mathbb{Z}\}$ of integers which are multiples of m is a subgroup of $(\mathbb{Z}, +)$, it is infinite if $m \neq 0$.
3. Given $n \geq 1$ the subset $H = \{1, r, \ldots, r^{n-1}\}$ of D_{2n} consisting of all powers of the rotation r is a subgroup of D_{2n} of cardinality n.
4. Given a group G and an element a of G the subset

$$H = \{a^n : n \in \mathbb{Z}\}$$

of G consisting of all integer powers of a is a subgroup, called the cyclic subgroup of G generated by a [note that $a^0 = aa^{-1} = e_G$]. The subgroup H is usually denoted $\langle a \rangle$, and a is called a generator of H.
5. The subset $\{e, (12)(34), (13)(24), (14)(23)\}$ is a subgroup of S_4.

Definition 1.3 (Order of an element).
Let G be a group and $a \in G$. Assume $\exists t \geq 1$ with $a^t = e_G$. We define the *order* of a by

$$\text{ord}(a) = \min \ \{n \geq 1 \ : \ a^n = e_G\}.$$

If $a^n \neq e_G, \forall n \geq 1$, we define $\text{ord}(a) = \infty$ and say a has infinite order.

Example 1.3 (Properties).

1. $\text{ord}(a) = 1$ if and only if $a = e_G$.
2. Let $0 \neq n \in \mathbb{Z}$. Then $\text{ord}(n) = \infty$, for otherwise $\exists t \geq 1$ with $tn = 0$ which is not possible.
3. If G is a group and $a \in G$ then $\text{ord}(a) = \infty$ is equivalent to saying that the elements $\{a^t : t \geq 1\} = \{a, a^2, \ldots, a^i, a^{i+1}, \ldots\}$ of G are all pairwise distinct: for otherwise if $i > j$ and $a^i = a^j$, then $a^{i-j} = e_G$ with $i - j \geq 1$, and hence a would have finite order. In particular, if G is a finite group, every element $a \in G$ has finite order. For otherwise G would contain the infinite set $\{a, a^2, \ldots, a^i, a^{i+1}, \ldots\}$, which is not possible since G is finite.
4. Let a be an element of a group G with finite order $\text{ord}(a) = k$. Then

$$H = \langle a \rangle = \{a^n : n \in \mathbb{Z}\} = \{e_G, a, a^2, \ldots, a^{k-1}\}$$

and $|H| = k$.

Proof. Let $n = tk + r$ with $0 \leq r < k$ (Euclidean division). Then $a^n = a^{tk+r} = a^{tk}.a^r = (a^k)^t a^r = (e_G)^t a^r = e_G a^r = a^r$. Also if $1 \leq j < i \leq k - 1$ and $a^i = a^j$ then $a^{i-j} = e_G$, hence $\text{ord}(a) \leq i - j < k$ which is a contradiction. Thus, the elements $\{e_G, a, a^2, \ldots, a^{k-1}\}$ are all pairwise distinct. □

5. Let $n \geq k \geq 2$ and $\sigma = (a_1 \cdots a_k) \in S_n$ a cycle of length k. Then $\text{ord}(\sigma) = k$. Further if $\sigma \in S_n$ then one can write $\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_m$: a product of disjoint cycles and $\text{ord}(\sigma) = \text{lcm}(\text{ord}(\tau_i))_{i=1}^m$ (will discuss examples in lectures).
6. Let $n \geq 1$ and $[k]_n \in \mathbf{Z}_n$. Then $\text{ord}([k]_n) = \frac{n}{\gcd(n,k)}$.

Proof. Let $d = \gcd(k, n)$. Write $dk' = k$ and $dn' = n$ with $\gcd(n', k') = 1$. We show $n' = \text{ord}([k]_n)$. First $n'[k]_n = [kn']_n = [dn'k']_n = [nk']_n = [0]_n$ so that $\beta = \text{ord}([k]_n) \leq n'$. Assume $\beta < n'$ then $[0] = \beta[k]_n = [\beta k]_n$ hence $n = dn'$ divides $\beta k = \beta dk'$, which implies n' divides $\beta k'$, the latter implies that n' divides β since $\gcd(n', k') = 1$ which contradicts $\beta < n'$. □

Definition 1.4 (Cyclic Groups).
A group G is called *cyclic* if $\exists a \in G$ such that

$$G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$$

equals the (sub)group generated by a. The element a is called a *generator* of G.

A generator of a cyclic group need not be unique, if a generates G then so does a^{-1} since $a^n = (a^{-n})^{-1} = (a^{-1})^{-n}$ for $n \in \mathbb{Z}$.

If $G = \langle a \rangle$ is cyclic, and $\text{ord}(a)$ is infinite, then

$$G = \{e, a, a^{-1}, a^2, a^{-2}, \ldots, a^n, a^{-n}, \ldots\}$$

is infinite, and $a^i \neq a^j$ if $i < j$ are integers.

If $G = \langle a \rangle$ is cyclic, and $\text{ord}(a) = k$ is finite, then $|G| = k$ and

$$G = \{e, a, \ldots, a^{k-1}\}.$$

Example 1.4.

1. $\mathbb{Z} = \langle 1 \rangle$ is an infinite cyclic group generated by 1. Indeed $\forall n \in \mathbb{Z}, n = n.1$. Also $\mathbb{Z} = \langle -1 \rangle$ since $n = (-n)(-1)$.
2. $\mathbf{Z}_n = \langle [1]_n \rangle$ is a cyclic group generated by $[1]_n$. Indeed $[h]_n = h[1]_n$ for all integers h. Further if $k \geq 1$ is an integer with $\gcd(k, n) = 1$ then $\mathbf{Z}_n = \langle [k]_n \rangle$ and $[k]_n$ is also a generator of $\mathbf{Z}_n$. Indeed $\text{ord}([k]_n) = \frac{n}{\gcd(k,n)} = n$ so that the subgroup $H = \langle [k]_n \rangle$ of $\mathbf{Z}_n$ generated by $[k]_n$ must equal $\mathbf{Z}_n$ as $H \subset \mathbf{Z}_n$ and they have the same cardinality n. For example $\mathbf{Z}_8 = \langle [3]_8 \rangle = \langle [5]_8 \rangle$.
3. If $x \in G$ has order n, then $\text{ord}(x^k) = \frac{n}{\gcd(n,k)}$ for each $k \geq 1$ (same proof as in example 2 above).
4. $H = \langle (1234) \rangle = \{e, (1234), (13)(42), (1432)\}$ is a cyclic subgroup of S_4 of cardinality 4. The subgroup $\{e, (12)(34), (13)(24), (14)(23)\}$ is not a cyclic subgroup of S_4 as it has no elements of order 4.
5. $\langle r \rangle = \{1, r, \ldots, r^{n-1}\}$ is a cyclic subgroup of cardinality n of D_{2n}. The subgroup $\{1, s, sr^2, r^2\}$ is a subgroup of D_8 which is not cyclic as it has no elements of order 4: $\text{ord}(r^2) = \text{ord}(s) = \text{ord}(sr^2) = 2$.

Definition 1.5 (Product of Groups).
Let $(G, \circ)$ and $(H, \star)$ be two groups. We define a new group $(G \times H, .)$, called the *product* of G and H, as follows

$$G \times H = \{(g, h) \ : \ g \in G, \ h \in H\}$$

is the set-theoretic product of G and H. The composition law $(.)$ is defined by

$$(g_1, h_1).(g_2, h_2) = (g_1 \circ g_2, h_1 \star h_2),$$

it is a group law on $G \times H$: associativity can be easily checked using associativity of $\circ$ and $\star$ (check this fact). The neutral element is (e_G, e_H):

$$(g,h).(e_G,e_H) = (g \circ e_G, h \star e_H) = (g,h),$$

and the inverse $(g,h)^{-1}$ of (g,h) is (g^{-1},h^{-1}):

$$(g,h).(g^{-1},h^{-1}) = (g \circ g^{-1}, h \star h^{-1}) = (e_G,e_H).$$

Lemma 1.1.
Let $(G,\circ)$ and $(H,\star)$ be groups. If G and H are abelian then so is $G \times H$. If both G and H are finite then so is $G \times H$ and $|G \times H| = |G||H|$.

Proof. Assume G, H are abelian, let $g_1, g_2 \in G$, and $h_1, h_2 \in H$. Then $(g_1,h_1).(g_2,h_2) = (g_1 \circ g_2, h_1 \star h_2) = (g_2 \circ g_1, h_2 \star h_1) = (g_2,h_2).(g_1,h_1)$, where we used the commutativity of $\circ$ and $\star$ for the second equality. If both G and H are finite then the number of elements of $G \times H$ is the number of pairs (g,h) with $g \in G$ and $h \in H$. There are $|G|$ possibilities for g, $|H|$ possibilities for h, which gives $|G||H|$ for the number of such pairs. □

Example 1.5.

1. $\mathbb{Z} \times \mathbb{Z} = \{(m,n) : m,n \in \mathbb{Z}\}$ is a group, we denote the group law by $+$ (and call it addition). Thus
$$(m,n) + (m',n') = (m+m', n+n'),$$
$(0,0)$ is the neutral element, and the inverse of (m,n) is $(-m,-n)$.
2. Given positive integers n and m: $\mathbf{Z}_n \times \mathbf{Z}_m = \{([k]_n,[r]_m) : k,r \in \mathbb{Z}\}$ is a group, we denote the group law by $+$. Thus
$$([k]_n,[r]_m) + ([k']_n,[r']_m) = ([k+k']_n,[r+r']_m),$$
$([0]_n,[0]_m)$ is the neutral element, and the inverse of $([k]_n,[r]_m)$ is $([-k]_n,[-r]_m)$. Further $\mathbf{Z}_n \times \mathbf{Z}_m$ is a finite group of cardinality mn.
3. $\mathbf{Z}_n \times \mathbf{Z} = \{([k]_n, r) : k,r \in \mathbb{Z}\}$ is a group, we denote the group law by $+$. Thus
$$([k]_n,r) + ([k']_n,r') = ([k+k']_n, r+r'),$$
$([0]_n, 0)$ is the neutral element, and the inverse of $([k]_n,r)$ is $([-k]_n,-r)$. Further $\mathbf{Z}_n \times \mathbf{Z}$ is an infinite group.

Definition 1.6 (Group Homomorphism).
Let $(G,\circ)$ and $(H,\star)$ be groups. A map $f : G \to H$ is called a *homomorphism* if

$$f(g_1 \circ g_2) = f(g_1) \star f(g_2), \quad \forall g_1, g_2 \in G.$$

In particular $f(e_G) = e_H$ (check this fact).

If in addition f is a bijection it is called an *isomorphism*.

Example 1.6.
Study examples 3,5 and 6 below.

1. The map $f : \mathbb{Z} \to \mathbf{Z}_n$ defined by $f(k) = [k]_n$ is a group homomorphism since

$$f(r+s) = [r+s]_n = [r]_n + [s]_n = f(r) + f(s).$$

It is surjective as $[k]_n = f(k)$ but is not injective: $f(k) = f(k+n) = [k]_n$.

2. Let $m \geq 0$ be an integer. The map $\phi : \mathbb{Z} \to \mathbb{Z}$ defined by $\phi(k) = mk$ is a group homomorphism:

$$\phi(k_1 + k_2) = m(k_1 + k_2) = mk_1 + mk_2 = \phi(k_1) + \phi(k_2).$$

This map is injective if $m \neq 0$: $\phi(k_1) = mk_1 = \phi(k_2) = mk_2 \implies m(k_1 - k_2) = 0 \implies k_1 - k_2 = 0 \implies k_1 = k_2$. The map ϕ is not surjective if $m \neq 1$: if m doesn't divide r then $r \neq \phi(k), \forall k \in \mathbb{Z}$.

3. Let $n > 1$ an integer. Recall that a permutation $\sigma \in S_n$ is *even* (resp. *odd*) if σ can be written as the product of an even (resp. odd) number of transpositions (ij). The map

$$\epsilon : S_n \to \mathbf{Z}_2$$

defined by $\epsilon(\sigma) = [0]_2$ if σ is even, and $\epsilon(\sigma) = [1]_2$ if σ is odd, is a group homomorphism (check this fact). It is surjective but is not injective if $n \geq 3$.

4. The map

$$f : D_{2n} \to \mathbf{Z}_2$$

defined by

$$f(s^i r^j) = [i]_2$$

is a group homomorphism. Suppose k is an odd integer then

$$f(s^i r^j . s^k r^l) = f(s^i r^j s r^l) = f(s^i s r^{n-j} r^l) = f(s^{i+1} r^{n-j+l}) = [i+1]_2 = [i+k]_2$$

$$[i]_2 + [k]_2 = f(s^i r^j) f(s^k r^l).$$

if k is even then

$$h(s^i r^j . s^k r^l) = h(s^i r^j r^l) = h(s^i r^{j+l}) = [i]_2 = [i]_2 + [0]_2 = h(s^i r^j) h(s^k r^l).$$

It is surjective: $f(r) = [0]_2$ and $f(s) = [1]_2$, but f is not injective: $f(s) = f(sr)$.

5. The map

$$f : D_4 \to \mathbf{Z}_2 \times \mathbf{Z}_2$$

defined by $f(1) = ([0]_2, [0]_2)$, $f(r) = ([1]_2, [0]_2)$, $f(s) = ([0]_2, [1]_2)$ and $f(sr) = ([1]_2, [1]_2)$ is an isomorphism (check that f is a homomorphism). The homomorphism f is bijective as it is injective and both groups have the same cardinality which equals 4, thus f is an isomorphism.

6. The map

$$h : D_{2n} \to \mathbf{Z}_n$$

defined by

$$h(s^i r^j) = [j]_n$$

is NOT a group homomorphism: if k is odd then

$$\begin{aligned} h(s^i r^j . s^k r^l) &= h(s^i s^k r^{n-j} r^l) = h(s^{i+k} r^{n-j+l}) \\ &= [n - j + l]_n = [-j + l]_n, \end{aligned}$$

which in general is different from $h(s^i r^j) h(s^k r^l) = [j]_n + [l]_n = [j + l]_n$.

Remark (Basic properties of homomorphisms).
Let $f : (G, \circ) \to (H, \star)$ be a group homomorphism.

1. Let $n \in \mathbb{Z}$ and $g \in G$ then $f(g^n) = f(g)^n$.
2. Suppose f is an isomorphism. Then the followings hold.
   (i) G is finite $\iff$ H is finite.
   (ii) G is abelian $\iff$ H is abelian.
   (iii) $\mathrm{ord}(g) = \mathrm{ord}(f(g)), \forall g \in G$.

Proof. See Problem Sheet 1, Q.6. □

Definition 1.7 (Image and Kernel of a Homomorphism).
Let

$$f : (G, \circ) \to (H, \star)$$

be a group homomorphism. We define the *image* of f to be

$$\mathrm{Im}(f) = \{h \in H \ : \ \exists g \in G, \ h = f(g)\}.$$

We define the *kernel* of f to be

$$\mathrm{Ker}(f) = \{g \in G \ : \ f(g) = e_H\}.$$

Remark (Basic properties of Image and Kernel).
Let $f : (G, \circ) \to (H, \star)$ be a group homomorphism. Then $\mathrm{Im}(f)$ is a subgroup of H and $\mathrm{Ker}(f)$ is a subgroup of G.

Proof. The fact that $\mathrm{Im}(f)$ is a subgroup of H has been proven in Year 1. We show $\mathrm{Ker}(f)$ is a subgroup of G. First, let $g \in \mathrm{Ker}(f)$, i.e. $f(g) = e_H$. Then $e_H = f(e_G) = f(g \circ g^{-1}) = f(g) \star f(g^{-1}) = e_H \star f(g^{-1}) = f(g^{-1})$ so that $g^{-1} \in \mathrm{Ker}(f)$. Second, let $g_1, g_2 \in \mathrm{Ker}(f)$. Then $f(g_1 \circ g_2) = f(g_1) \star f(g_2) = e_H \star e_H = e_H$ and $g_1 \circ g_2 \in \mathrm{Ker}(f)$. □

Example 1.7.

1. The group homomorphism $f : \mathbb{Z} \to \mathbf{Z}_n$ in Example 1.6 has image $\mathrm{Im}(f) = \mathbf{Z}_n$ and kernel $\mathrm{Ker}(f) = n\mathbb{Z}$.
2. The group homomorphism $g : \mathbb{Z} \to \mathbb{Z}$ in Example 1.6 has image $\mathrm{Im}(f) = m\mathbb{Z}$ and kernel $\mathrm{Ker}(f) = \{0\}$ if $m \neq 0$.
3. The group homomorphism $\epsilon : S_n \to \mathbf{Z}_2$ in Example 1.6, for $n > 1$, has image $\mathrm{Im}(f) = \mathbf{Z}_2$ and kernel $\mathrm{Ker}(f) = A_n$: the alternating subgroup of S_n which consists of all even permutations.
4. The group homomorphism $\tau : D_{2n} \to \mathbf{Z}_2$ in Example 1.6 has image $\mathrm{Im}(f) = \mathbf{Z}_2$ and kernel $\mathrm{Ker}(f) = \langle r \rangle = \{1, r, \cdots, r^{n-1}\}$.

Proposition 1.2.
Let $f : (G, \circ) \to (H, \star)$ be a group homomorphism. Then

1. *f is surjective if and only if $Im(f) = H$.*
2. *f is injective if and only if* $\mathrm{Ker}(f) = \{e_G\}$.

Proof. Exercise (compare with the proof of Lemma 6.1 in §6). □